Formcraft Studio

REVIT FAMILIES

Healthcare Family Catalogue

A sample of eleven hospital families from the 500+ family pack: theatres, critical care, wards, imaging, pharmacy, laboratories, standby power and fire safety. Built to LOD 300, LOD 400 and LOD 400+ Render, driven by parameters, connected for MEP and carrying the data a hospital project needs.

LOD 300 · 400 · 400+ Render

Flex-tested

MEP connectors

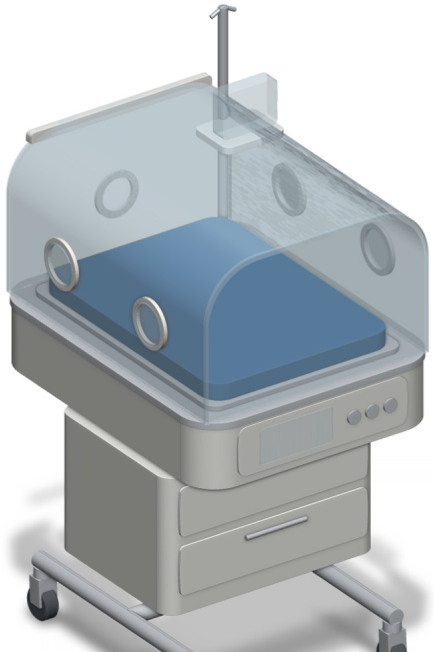
Clearance zones

Keynotes + IFC









THE FORMCRAFT STANDARD

How a Formcraft family is built

Every family in this catalogue is made the same way, from a written recipe, so each one behaves predictably in your project and can be checked before it reaches you.

01

Driven by reference planes

Every solid sits on labelled reference planes. Width, Depth and Height drive all of them through formulas, so a new size is a new type, not a remodel. Hardware keeps its real size and moves with the body.

02

Flex-tested before release

Each family is flexed at 80% and 125% of every size. It passes only when no part is left behind, ends up outside, or comes loose. Every type is regenerated in Revit and checked again.

03

LOD 300 detail at Fine, light at Coarse

Fine shows the full product: hinges, castors, keys, gaskets. Medium drops small hardware. Coarse shows a clean envelope, so large plans stay fast and readable.

04

Connected for MEP

Pipe, duct, power and data connectors sit on their own labelled planes, with systems set: cold and hot water, waste, medical gases, supply air, 230/400 V power, ELV and data.

05

Clearances that never print

Patient space, maintenance, access and door approach zones are modelled on their own subcategories, sized by parameter and based on HBN, HTM and BS guidance. Turn them off for drawings.

06

Data for the whole project

Keynote on every type, IFC class, cost unit, power risk grade and supply source, mounting height, host type, legend sheet and clearance basis, all as shared parameters you can schedule.

WHY THE SHAPES ARE SIMPLE

Light by design

Think of an architect's massing model: every block the right size, in the right place, with the right connections, but no wasted detail. Formcraft families draw rounded and complex forms as simple solids, true to size, and carry the detail in their data instead.

01 **Fast models**

A hospital model holds thousands of families. Simple solids keep each file light, 0.6 to 1.5 MB for a standard family, so views open, sync and print quickly for the whole team.

02 **True where it counts**

Overall size, clearances, mounting heights and service connections are exact. A designer coordinating a room needs those, not the curve of a handle.

03 **Every part moves with the sliders**

Because each solid sits on labelled reference planes, changing Width, Depth or Height resizes the whole product cleanly. That is what the flex test checks.

OPTION

Prefer the true shape? Ask for detailed 3D

Any family in your set can also come as a detailed model.

- True curves and profiles, close to the product photograph
- Sizes set per type rather than by sliders
- Best for renders, client presentations and close-up coordination, at 1.4 to 2.6 MB per family as built
- Mix and match: detailed versions for the showpieces, light ones for everything else

PRODUCT MATCHING SERVICE

Your products, modelled to this standard

The families in this catalogue are generic: typical sizes, no brand. Send us your product catalogue and we will build your exact range in Revit, with the same detail and data as these pages.

01 **Send us your catalogue**
PDF datasheets, dimension drawings or a product list. One product or a whole range.

02 **We match each product**
Each product is built on the Formcraft family that fits it, so it keeps the same parametric skeleton, flex test and detail as this catalogue.

03 **Your models become types**
One type per model number, with your sizes, product codes, finishes, and the loads, flows and connections from your data sheets.

04 **Checked against your drawings**
Every family is compared side by side with your dimension drawings and flex-tested before you see it.

05 **Delivered as a set**
Revit families, a keynote legend and a catalogue like this one for your own range, ready to send to designers.

Who it is for

- Medical equipment suppliers who want designers to specify their products
- Hospital projects that have chosen their equipment and need it in the model
- Manufacturers who want their range listed in Revit libraries

Start with one product

Send one data sheet and we model it as a trial, so you can check it in your own project first.

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THE FULL CATALOGUE

Buy the full set, get support with it

This catalogue shows 11 of the Formcraft Healthcare pack. The full pack, 500+ hospital families across every discipline, comes with 1 month of Formcraft support, covering up to 500 fixes or changes: about one per family.

Made yours

We brand the families for your practice: your company in the family data, your naming and your details on the keynote legends.

Fixed and sent back

If a family fails in your project, send it to us. We fix it fast and send the corrected family back.

Upgrades included

When we improve a family in the catalogue, you receive the new version.

If a family fails

01 Send the family and a note of what went wrong: the Revit warning or a screenshot.

02 We open it, fix it and run the flex test again.

03 You get the corrected .rfa back with a short note of what changed.

Support runs 1 month from purchase, up to 500 fixes or changes. Renewal available.

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PRICES

Choose your set

Single family

$2 per family

≈ UGX 8,300

- ✓ Any family from the catalogue
- ✓ Light, parametric, flex-tested
- ✓ Types, data, keynote and connectors

Full catalogue

$850 500+ families · save $150

≈ UGX 3.5 million

- ✓ The whole Healthcare pack, every discipline
- ✓ Keynote legends for each discipline
- ✓ Types, data and MEP connectors throughout

Recommended

Catalogue + support

$1,200 500+ families, 1 month support

≈ UGX 5.0 million

- ✓ Everything in the full catalogue
- ✓ Families branded for your practice
- ✓ Failed families fixed fast and sent back
- ✓ 1 month support, up to 500 fixes or changes
- ✓ Upgrades as families improve
- ✓ Detailed 3D versions of selected families

Billed in US dollars. Shilling amounts are a guide at 1 USD ≈ 4,130 UGX (16 Oct 2026). Your own products modelled to this standard (page 4) are quoted per range.

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THE SAMPLE SET

What you receive

The family files

11 Revit 2025 families, each as two .rfa files: the LOD 300 family and the detailed family (LOD 400 and 400+ Render), named to the Formcraft standard and grouped by discipline.

Types ready to use

Every family carries its standard sizes as types. Pick a type, or change the size parameters yourself.

Tested for flexing

Each family passed a flex test: Width, Depth and Height pushed to 80% and 125%, with no part left behind or floating.

Data on board

Keynote, IFC class, cost unit, clearance rule, power grade and standards in shared parameters, ready for schedules.

Keynote legends

A legend per discipline that ties each keynote code to its description, for drawings and specifications.

Connectors

Power, data, water, drainage and medical gas connectors where the real item needs them, so systems and loads work.

How to use them

01 Load the family (Insert › Load Family), or drag it from the folder into the project.

02 Place it on the right host: floor, wall face, ceiling or a wall for doors and windows.

03 Choose a type, then adjust the size and visibility parameters in Properties.

04 Tag it with the keynote tag and schedule it with the shared parameters.

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THE SAMPLE

Eleven families, three levels each

The most detailed families from the hospital pack, chosen to show the range: each shown as the parametric family (LOD 300), the detailed family (LOD 400, faceted, not for rendering) and its LOD 400+ Render version with the smooth Rhino shape. Doors and windows included.



Detailed ICU electric bed

Medical equipment · p. 11

300 · 400 · 400+ Render



Detailed ceiling operating light

Electrical · p. 12

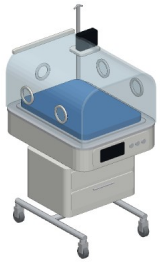
300 · 400 · 400+ Render



Detailed anaesthesia machine

Medical equipment · p. 13

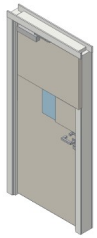
300 · 400 · 400+ Render



Detailed neonatal incubator

Medical equipment · p. 14

300 · 400 · 400+ Render



Hygienic flush door

Architecture · p. 15

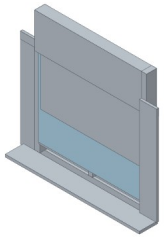
300 · 400 · 400+ Render



Lead-lined X-ray door

Architecture · p. 16

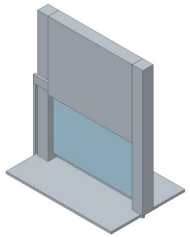
300 · 400 · 400+ Render



Nurse observation window

Architecture · p. 17

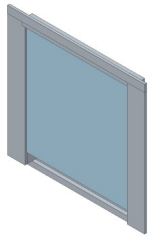
300 · 400 · 400+ Render



Pharmacy dispensing hatch

Architecture · p. 18

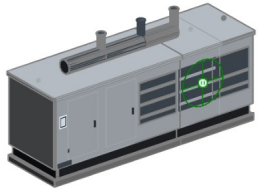
300 · 400 · 400+ Render



Laboratory window

Architecture · p. 19

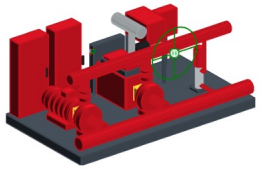
300 · 400 · 400+ Render



Diesel standby generator

Electrical · p. 20

300 · 400 · 400+ Render



Sprinkler fire pump set

Fire protection · p. 21

300 · 400 · 400+ Render

11
Revit 2025 families (.rfa)

≈ 8.5 MB
LOD 300, all eleven: about 0.55–1.1 MB each

2 files
per family: LOD 300, and one detailed file holding LOD 400 and 400+ Render (0.7–2.6 MB)

1 zip
Formcraft_Healthcare_Sample11.zip: this catalogue and all 22 files

IN YOUR PROJECT

What every family gives you

Load any of the eleven into a Revit 2025 project and try these yourself: tag it, schedule it, put it in a legend, connect it, switch the detail level.

01 Parametric types

Width, Depth and Height drive labelled reference planes, so a new size is a new type, not a remodel. Every family is flex-tested at 80% and 125% before release.

02 Three levels of detail

A clean box at Coarse for big plans, the full body at Medium, the finest parts at Fine. The four detailed families go up to a smooth LOD 400+ Render version.

03 Materials you can tag

Every part carries a Formcraft material with its keynote and cost code, so material tags and material takeoff schedules read real values.

04 Keynotes on every type

Each type carries its keynote, so keynote tags and keynote legends work straight away, by element or by material.

05 Tags and schedules

Shared parameters by GUID: description, cost unit, power risk grade, supply source, mounting height and IFC class, all schedulable and taggable.

06 Legend views

Every family places as a legend component in plan, front and side. The Formcraft tools build a keynote legend and type schedule per discipline.

07 Connected for MEP

Power, data, water, drainage, medical gases and ductwork connectors, with their systems set, so the families join real MEP systems.

08 Clearance zones

Patient space, maintenance, access and door approach zones on their own subcategories, based on HBN, HTM and BS guidance. Turn them off for drawings.

09 Hosted correctly

Wall and ceiling items are face-based: they follow their host, also in linked models. Each was placed, moved with its host and checked.

10 IFC and handover

Each type exports to its IFC class, with the asset and cost fields filled for handover and quantity surveying.

11 Previews

Each family shows its picture in Revit's Load Family dialog and in Windows Explorer, so a library can be browsed by eye.

12 Light files

0.6 to 1.5 MB for a standard family. A LOD 400+ Render family measures 1.4 to 2.6 MB as built: one file holds the LOD 400 body and the smooth Rhino parts, and Revit switches between them by detail level.

HONEST ABOUT IT

Strengths and limits

What the system does well today, and what to keep in mind while you evaluate it. Each limit has a way round it or a fix on the way.

Strengths

+ One standard for 500+ families: the same planes, names, data and checks everywhere, so the library behaves predictably.
+ Truly parametric: sizes change by type through labelled planes, and every family passes a flex test before release.
+ Ready for engineers: MEP connectors with systems, including medical gases, plus power risk grades and supply sources.
+ Ready for quantity surveyors and handover: keynotes, materials, cost units and IFC classes on every type.
+ Health guidance built in: clearance zones from HBN, HTM and BS, sized by parameter.
+ Light where it matters: a box at Coarse, full detail only at Fine, 0.6 to 1.5 MB per file.
+ Made from written recipes, so a fix or a change to one family is quick and consistent: the basis of our support.

Limits to know

– Generic products: sizes and ratings are typical. Manufacturer and model fields are left blank; we can model your chosen products to match.
– Curved detail is fixed in shape per type. Straight parts stretch with sliders; curved parts change size by type.
– The four detailed families, the generator and the fire pump are built at LOD 400+ Render (smooth Rhino parts in the same file as the LOD 400 body, 1.4 to 2.6 MB each, measured). The doors and windows are built at LOD 400 with true-shape hardware. Each page shows which levels are ready.
– Revit 2025 files: they open in Revit 2025 or later, not in older versions.
– Revit’s Medical Equipment category cannot join pipe systems, so medical gas and water reach those items through a nested service point.
– Clearance zones show until you switch their subcategory off in the view; a non-printing setting is coming in the next kit.
– Not yet signed off by discipline professionals: this is an evaluation sample, and your review is part of that sign-off.

DETAILED 3D · MEDICAL EQUIPMENT · CRITICAL CARE

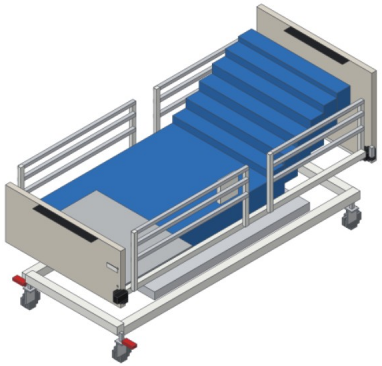
Detailed ICU electric bed

FMC_I_MedEq_Bed_ICU_Electric_Detailed

One family · LOD 300 · 400 · 400+ Render

LOD 300

PARAMETRIC



0.91 MB file, measured

The standard family: fully parametric, 234 parameters, 212 labelled dimensions, flexes with sliders. For planning and design development.

LOD 400

FACETED · NOT FOR RENDER



1.46 MB file, measured

Native Revit body with types, data, connectors and clearance. Flat-faced, not smooth: for design and documentation, not for rendering.

LOD 400+ Render

SMOOTH



2.61 MB file, measured

One file, built by script in 4.9 min (1,458 steps): 65 smooth parts modelled in Rhino, 69 native solids, 10 kit materials. Revit shows a box at Coarse, the LOD 400 body at Medium and the smooth parts at Fine. For presentation and rendering.

The ICU bed at three levels of detail in one family: contoured mattress with the backrest raised 30°, rounded head and foot boards with grips, tubular split side rails with control panels, lift columns, castors and the red CPR release.

Category · host	Medical Equipment · floor (level based)
Types	1020 × 2200 standard
Connectors	Power · data (nurse call)
Clearance	Patient space · HBN 04-01, HBN 09-03
Data	Keynote F.MED_00 · IfcMedicalDevice · cost unit nr

DETAILED 3D · ELECTRICAL · THEATRES

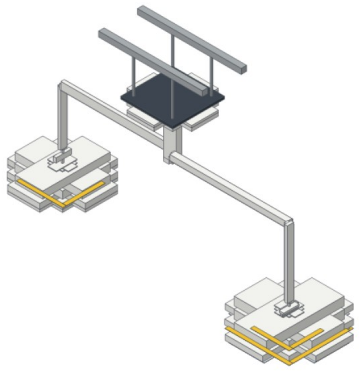
Detailed ceiling operating light

FMC_E_LightFixture_OperatingLight_Ceiling_Detailed

One family · LOD 300 · 400 · 400+ Render

LOD 300

PARAMETRIC



0.89 MB file, measured

The standard family: fully parametric, 273 parameters, 212 labelled dimensions, flexes with sliders. For planning and design development.

LOD 400

FACETED · NOT FOR RENDER



1.01 MB file, measured

Native Revit body with types, data, connectors and clearance. Flat-faced, not smooth: for design and documentation, not for rendering.

LOD 400+ Render

SMOOTH



1.43 MB file, measured

One file, built by script in 4.0 min (1,319 steps): 33 smooth parts modelled in Rhino, 38 native solids, 7 kit materials. Revit shows a box at Coarse, the LOD 400 body at Medium and the smooth parts at Fine. For presentation and rendering.

Ceiling-mounted surgical light at three levels of detail: domed light heads with LED modules behind a frosted lens, sterile centre handles, curved spring arms and yokes on a central axis. The second head switches on with the double-dome type.

Category · host	Lighting Fixtures · ceiling face (face-based)
Types	700 single · 700 + 600 double dome
Connectors	Power
Clearance	Maintenance zone · CIBSE LG2, HTM 03-01
Data	Keynote E.LED_13 · IfcLightFixture · cost unit nr

DETAILED 3D · MEDICAL EQUIPMENT · THEATRES

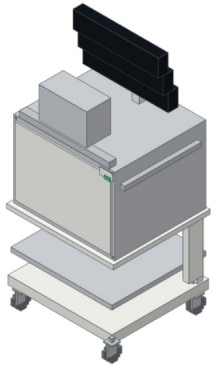
Detailed anaesthesia machine

FMC_I_MedEq_Machine_Anaesthesia_Detailed

One family · LOD 300 · 400 · 400+ Render

LOD 300

PARAMETRIC



0.75 MB file, measured

The standard family: fully parametric, 149 parameters, 138 labelled dimensions, flexes with sliders. For planning and design development.

LOD 400

FACETED · NOT FOR RENDER



1.31 MB file, measured

Native Revit body with types, data, connectors and clearance. Flat-faced, not smooth: for design and documentation, not for rendering.

LOD 400+ Render

SMOOTH



2.20 MB file, measured

One file, built by script in 3.3 min (1,156 steps): 65 smooth parts modelled in Rhino, 73 native solids, 8 kit materials. Revit shows a box at Coarse, the LOD 400 body at Medium and the smooth parts at Fine. For presentation and rendering.

Anaesthesia workstation at three levels of detail: sloped, rounded tower, flowmeter bank and round gauges, ventilator bellows, CO2 absorber, breathing hoses, vaporisers, patient monitor on a curved arm, drawers and reserve cylinders.

Category · host	Medical Equipment · floor (level based)
Types	770 × 690 standard
Connectors	Power · medical gas hoses ×5
Clearance	Operation zone · HBN 26
Data	Keynote Z.MEQ_01 · IfcMedicalDevice · cost unit nr

DETAILED 3D · MEDICAL EQUIPMENT · NEONATAL

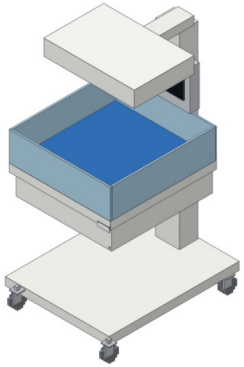
Detailed neonatal incubator

FMC_I_MedEq_Incubator_Neonatal_Detailed

One family · LOD 300 · 400 · 400+ Render

LOD 300

PARAMETRIC

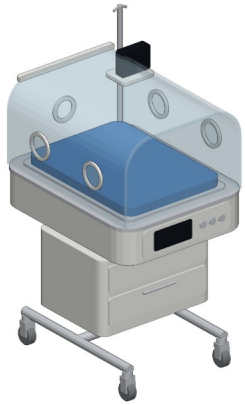


0.70 MB file, measured

The standard family: fully parametric, 130 parameters, 112 labelled dimensions, flexes with sliders. For planning and design development.

LOD 400

FACETED · NOT FOR RENDER



0.94 MB file, measured

Native Revit body with types, data, connectors and clearance. Flat-faced, not smooth: for design and documentation, not for rendering.

LOD 400+ Render

SMOOTH



1.40 MB file, measured

One file, built by script in 3.1 min (905 steps): 47 smooth parts modelled in Rhino, 52 native solids, 8 kit materials. Revit shows a box at Coarse, the LOD 400 body at Medium and the smooth parts at Fine. For presentation and rendering.

Closed infant incubator at three levels of detail: arched, double-walled clear canopy, round hand ports and an iris port, mattress tray, control module with display, drawer stand, IV pole with monitor shelf and a castored H-base.

Category · host	Medical Equipment · floor (level based)
Types	780 × 990 standard
Connectors	Power · medical gas hoses ×2
Clearance	Patient space · HBN 04-01, HBN 09-03
Data	Keynote F.MED_07 · IfcMedicalDevice · cost unit nr

ARCHITECTURE · DOORS

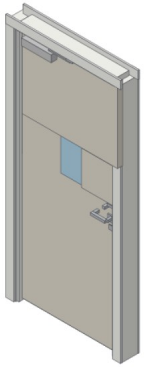
Hygienic flush door

FMC_A_Door_FlushSingle_Hygienic

LOD 300 · 400 · 400+ Render

LOD 300

PARAMETRIC



0.71 MB file, measured

The standard family: fully parametric, 110 parameters, 102 labelled dimensions, flexes with sliders. For planning and design development.

LOD 400

FACETED · NOT FOR RENDER



1.09 MB file, measured

The detailed family: true outlines and the small parts (hinges, handles, seals, grilles). Flat-faced, not smooth: for documentation, not for rendering.

LOD 400+ Render

SMOOTH



1.09 MB file, measured

The same file as LOD 400, built by script in 7.7 min (860 steps): 35 solids, 9 nested hardware parts, 8 kit materials, Open toggle and clearance. No separate Rhino shapes are needed: its curved parts are already true shape.

Single HPL-faced flush door for clinical corridors and rooms, with hygienic seals, a vision panel and kick plates. Built in the same scheme as all Formcraft doors: frame, stops, leaf, lipping, closer, lever set and hinges each on their own planes.

Category · host	Doors · wall-hosted opening (Metric Door template)
Types	0926×2100, 1026×2100, 1126×2100 single leaf
Clearance	Door approach zones both faces · plan swing on planes
Performance	Acoustic 30 dB · lead lining available as a type option
Data	Keynote D.HOS_00 · IfcDoor · cost unit nr

ARCHITECTURE · IMAGING

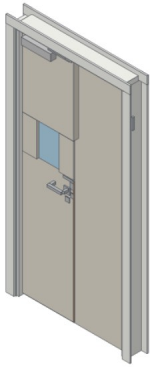
Lead-lined X-ray door

FMC_A_Door_LeadLined_Xray

LOD 300 · 400 · 400+ Render

LOD 300

PARAMETRIC



0.70 MB file, measured

The standard family: fully parametric, 109 parameters, 102 labelled dimensions, flexes with sliders. For planning and design development.

LOD 400

FACETED · NOT FOR RENDER

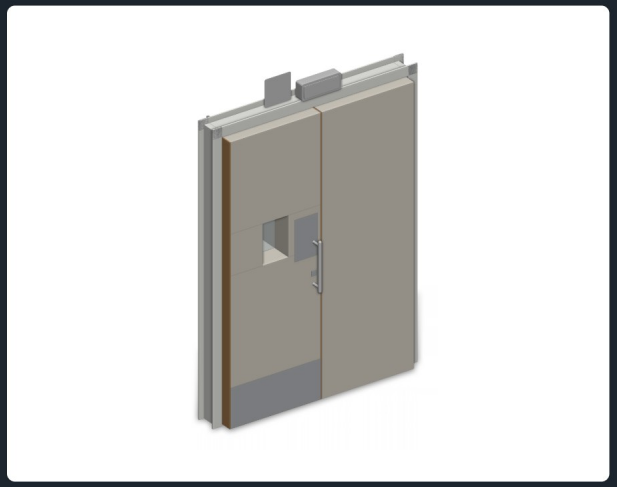


1.35 MB file, measured

The detailed family: true outlines and the small parts (hinges, handles, seals, grilles). Flat-faced, not smooth: for documentation, not for rendering.

LOD 400+ Render

SMOOTH



1.35 MB file, measured

The same file as LOD 400, built by script in 8.1 min (859 steps): 35 solids, 10 nested hardware parts, 8 kit materials, Open toggle and clearance. No separate Rhino shapes are needed: its curved parts are already true shape.

Radiation shielding door with lead lining in leaf and frame, overlapping stops, lead-glass vision panel, kick plates and push and pull hardware.

Category · host	Doors · wall hosted
Types	1026 × 2100 · 1200 × 2100 single · 1500 × 2100 double
Shielding	Lead lining (thickness set by the radiation adviser)
Clearance	Door approach zones both sides
Standards	IRR17 guidance · BS 8300-2

ARCHITECTURE · WINDOWS

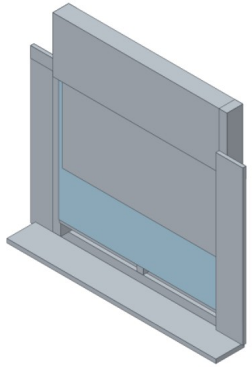
Nurse observation window

FMC_A_Window_Observation_Internal_Nurse

LOD 300 · 400 · 400+ Render

LOD 300

PARAMETRIC

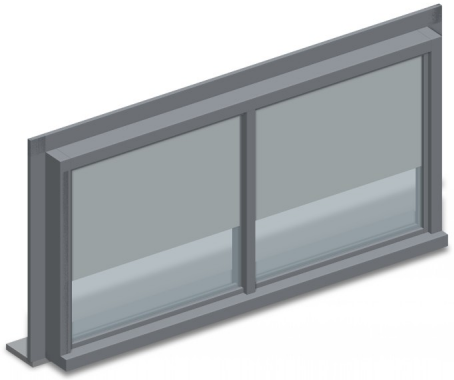


0.60 MB file, measured

The standard family: fully parametric, 48 parameters, 42 labelled dimensions, flexes with sliders. For planning and design development.

LOD 400

FACETED · NOT FOR RENDER

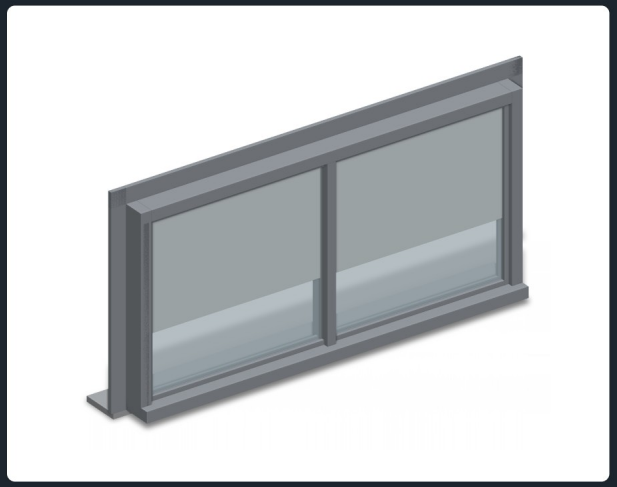


0.67 MB file, measured

The detailed family: true outlines and the small parts (hinges, handles, seals, grilles). Flat-faced, not smooth: for documentation, not for rendering.

LOD 400+ Render

SMOOTH



0.67 MB file, measured

The same file as LOD 400, built by script in 2.6 min (399 steps): 28 solids, 1 nested hardware part, 6 kit materials, Open toggle and clearance. No separate Rhino shapes are needed: its curved parts are already true shape.

Internal glazed observation panel between a nurse base and a ward or isolation room. Double glazing with an integral blind between the panes and a room-side control knob, so privacy needs no curtain to clean.

Category · host	Windows · wall-hosted opening (Metric Window template)
Types	1200×900 and 1800×900, fixed with side-hung option
Full detail	Frame, beads, two 6 mm panes, integral blind and knob
Data	Keynote G.FIX_02 · IfcWindow · cost unit nr
Materials	Aluminium, clear glass, stainless board

ARCHITECTURE · PHARMACY

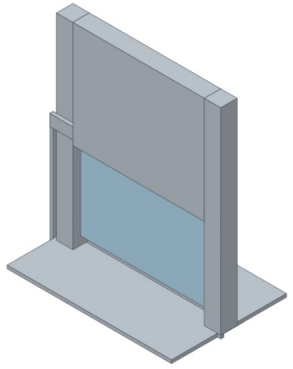
Pharmacy dispensing hatch

FMC_A_Window_Hatch_Pharmacy_Dispensing

LOD 300 · 400 · 400+ Render

LOD 300

PARAMETRIC

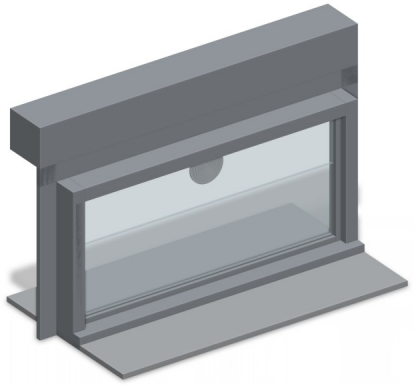


0.58 MB file, measured

The standard family: fully parametric, 43 parameters, 30 labelled dimensions, flexes with sliders. For planning and design development.

LOD 400

FACETED · NOT FOR RENDER

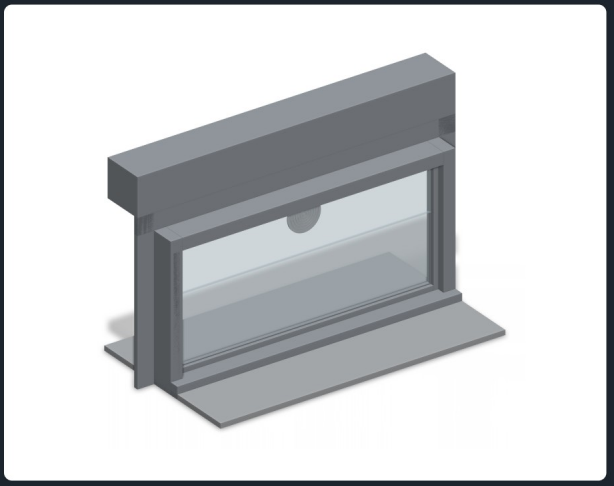


0.69 MB file, measured

The detailed family: true outlines and the small parts (hinges, handles, seals, grilles). Flat-faced, not smooth: for documentation, not for rendering.

LOD 400+ Render

SMOOTH



0.69 MB file, measured

The same file as LOD 400, built by script in 1.2 min (328 steps): 24 solids, 1 nested hardware part, 6 kit materials, Open toggle and clearance. No separate Rhino shapes are needed: its curved parts are already true shape.

Dispensing hatch for the pharmacy wall: frame, glazing beads on both faces, sill, window board, architraves and a stainless transaction shelf that can be switched off.

Category · host	Windows · wall hosted
Types	900 × 600 dispensing · 1200 × 600 dispensing
Options	Show Shelf (transaction counter)
Materials	Powder-coated aluminium, clear glass, brushed stainless
Clearance	Hatch user zone 900 deep · BS 8213, HBN 00-10

ARCHITECTURE · LABORATORIES

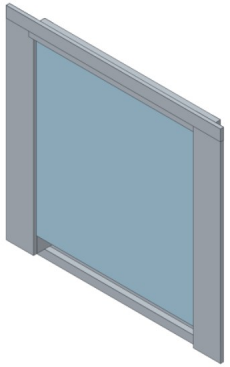
Laboratory window

FMC_A_Window_Laboratory_Fixed

LOD 300 · 400 · 400+ Render

LOD 300

PARAMETRIC

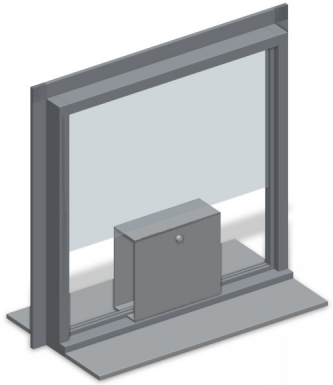


0.55 MB file, measured

The standard family: fully parametric, 30 labelled dimensions, 2 types, flexes with sliders. For planning and design development.

LOD 400

FACETED · NOT FOR RENDER

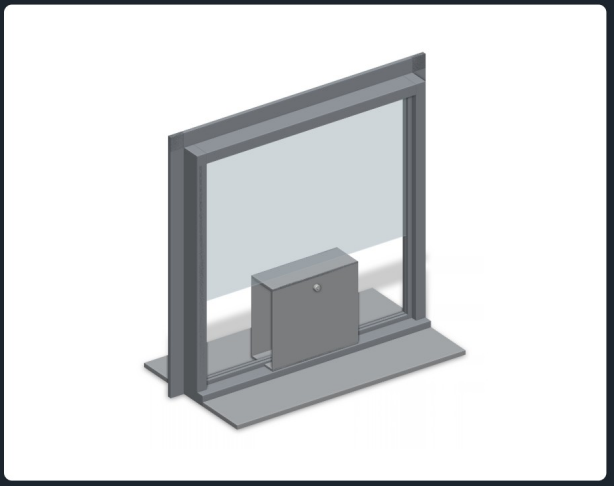


0.66 MB file, measured

The detailed family: true outlines and the small parts (hinges, handles, seals, grilles). Flat-faced, not smooth: for documentation, not for rendering.

LOD 400+ Render

SMOOTH



0.66 MB file, measured

The same file as LOD 400, built by script in 1.9 min (386 steps): 27 solids, 2 nested hardware parts, 5 kit materials, Open toggle and clearance. No separate Rhino shapes are needed: its curved parts are already true shape.

Fixed laboratory window with a sealed, flush-glazed frame and coved, ledge-free reveals so the room can be cleaned and kept sealed, with an optional pass-through hatch type for specimens.

Category · host	Windows · wall-hosted opening
Types	1200 × 1200 fixed · 1200 × 1200 with pass-through hatch
Clearance	Access zone in front of the hatch
Data	Keynote G.FIX_06 (proposed) · IfcWindow · cost unit nr

ELECTRICAL · STANDBY POWER

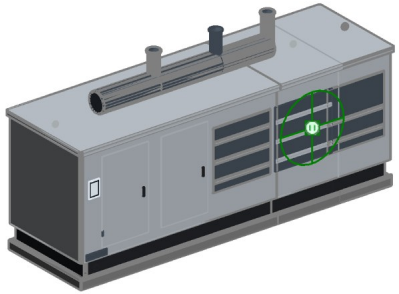
Diesel standby generator

FMC_E_EQ_Generator_Diesel_Standby

LOD 300 · 400 · 400+ Render

LOD 300

PARAMETRIC

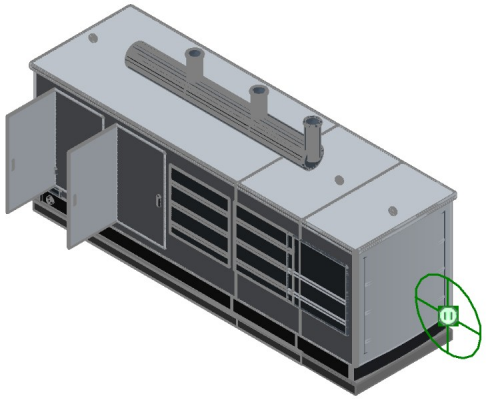


1.06 MB file, measured

The standard family: fully parametric, 128 parameters, 51 labelled dimensions, flexes with sliders. For planning and design development.

LOD 400

FACETED · NOT FOR RENDER

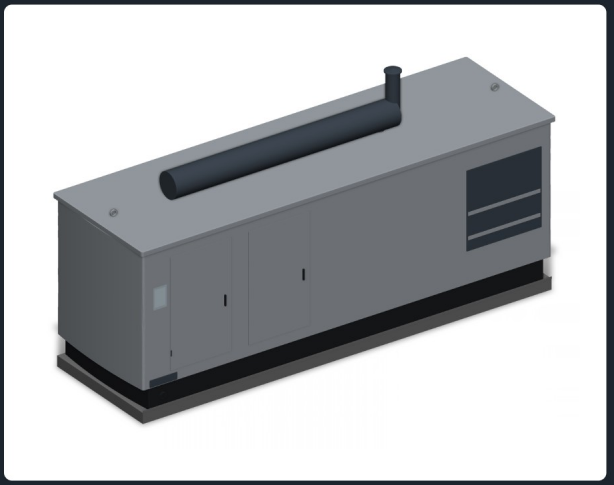


1.96 MB file, measured

The detailed family: true outlines and the small parts (hinges, handles, seals, grilles). Flat-faced, not smooth: for documentation, not for rendering.

LOD 400+ Render

SMOOTH



1.96 MB file, measured

One file, built by script in 11.1 min (993 steps): 44 smooth parts modelled in Rhino, 73 native solids, 9 kit materials. Revit shows a box at Coarse, the LOD 400 body at Medium and the smooth parts at Fine. For presentation and rendering.

Diesel standby generating set in an acoustic canopy on a base day tank, 415 V three-phase, with radiator end louvre, roof silencer and exhaust, control panel and access doors.

Category · host	Electrical Equipment · floor (level based)
Types	250 kVA · 500 kVA · 800 kVA canopy
Connectors	Power (415 V three-phase)
Clearance	Maintenance zone around the canopy
Materials	Powder-coated and galvanised steel, glass display

FIRE PROTECTION · PUMPS

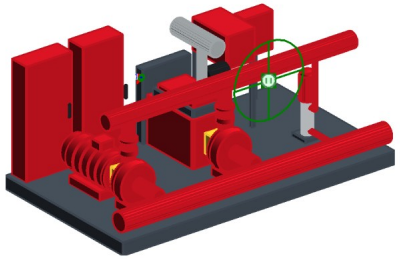
Sprinkler fire pump set

FMC_FP_MechEq_FirePump_Set

LOD 300 · 400 · 400+ Render

LOD 300

PARAMETRIC

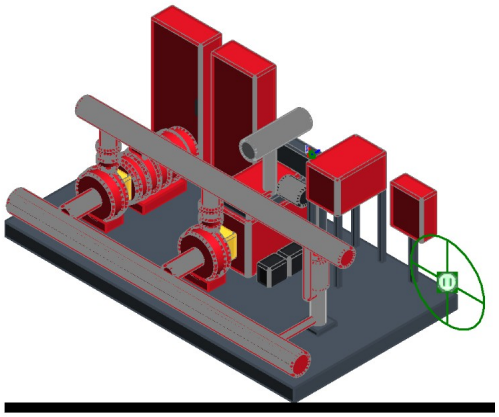


1.03 MB file, measured

The standard family: fully parametric, 131 parameters, 58 labelled dimensions, flexes with sliders. For planning and design development.

LOD 400

FACETED · NOT FOR RENDER

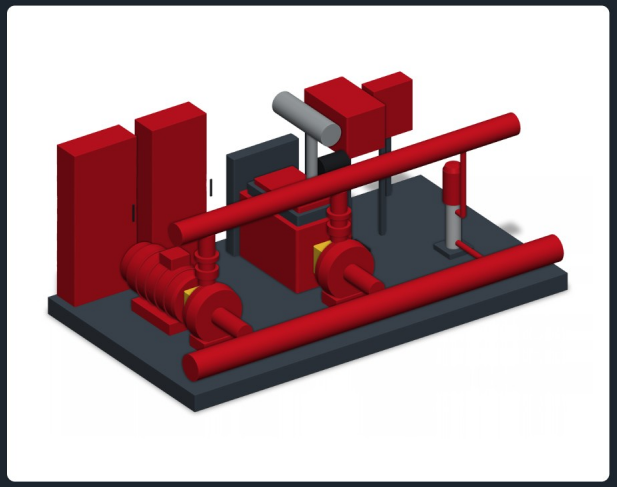


1.55 MB file, measured

The detailed family: true outlines and the small parts (hinges, handles, seals, grilles). Flat-faced, not smooth: for documentation, not for rendering.

LOD 400+ Render

SMOOTH



1.55 MB file, measured

One file, built by script in 6.2 min (742 steps): 40 smooth parts modelled in Rhino, 57 native solids, 10 kit materials. Revit shows a box at Coarse, the LOD 400 body at Medium and the smooth parts at Fine. For presentation and rendering.

Packaged sprinkler pump set on one skid: electric main pump, diesel standby pump with radiator, exhaust and day tank, jockey pump, suction and discharge headers and three controllers.

Category · host	Mechanical Equipment · floor (level based)
Types	3-pump set: electric, diesel, jockey
Connectors	Fire water (wet) pipe · power
Clearance	Maintenance zone around the skid
Materials	Red enamel, galvanised pipe, stainless, steel

All 594 hospital families

Bold names with a page number are shown in this catalogue. Every family carries types, data, keynotes and its service connectors.

Architecture 120

Door · Access Controlled Staff
Door · Access Panel Riser
Door · Acoustic 42 dB
Door · Automatic Sliding Entrance
Door · Automatic Sliding Isolation
Door · Cleanroom Pharmacy
Door · Cold Room Insulated
Door · Double Action Theatre
Door · Double Bed Passage FD60
Door · Fire Rated FD30 Ward
Door · Fire Rated FD60 Corridor
Door · Flush Single Hygienic p.15
Door · Flush To Wall Panel
Door · Glazed Metal Frame ICU
Door · Hatch Linen Return
Door · Hermetic Sliding Theatre
Door · Isolation Anteroom
Door · Lead Lined Sliding CT
Door · Lead Lined Xray p.16
Door · Leaf And Half Bed Passage
Door · Lift Landing Door Bed
Door · Louvred Steel Plant
Door · Mortuary Body Passage
Door · Nursery Glazed Vision
Door · Pair Lift Lobby FD60 Hold Open
Door · Patch Fitting Glass Entrance
Door · Psychiatric Anti Ligature
Door · RF Shielded MRI
Door · Screen Sidelights Transom
Door · Shower Bifold Accessible
Door · Single Sidelight Consult
Door · Sliding Manual Off Wall
Door · Sliding Pocket
Door · Sluice Single
Door · Smoke Cross Corridor Hold Open
Door · Stainless Full
Door · Steel General Purpose
Door · Stretcher Bifold Store
Door · Toilet Accessible Outward Opening
Door · Toilet Anti Barricade
Door · Transom Glazed
Door · Ward Single Observation
Bollard Ambulance Bay
Canopy Ambulance Bay
Clearance Bed Space
Clearance Door Approach
Clearance Maintenance
Clearance Turning Circle
Partition Cubicle Glazed ICU
Ramp Stretcher External
Waste Store Clinical Cage
Profile · Crash Rail Section
Profile · Handrail Hospital Section
Profile · Hygienic Coving Floor
Profile · Nosing Stair Anti Slip
AED Cabinet
Anti Ligature Fittings Set
Baby Change Unit
Backrest Pad WC
Bedhead Trunking Horizontal
Bedpan Rack
Changing Places Unit
Curtain Track Bed Cubicle
Decontamination Shower Mass
Dispenser Glove Apron
Dispenser Paper Towel Bin
Dispenser Sanitiser Auto
Dispenser Soap Auto
Door Edge Protector
Door Protection Plate
Enclosure Fire Rated Cabinet
Grab Rail Drop Down
Grab Rail L Shape
Grab Rail Straight
Hand Hygiene Station
IV Track Ceiling
Lift Bed Passenger
Lift Dumbwaiter
Lift Goods Service
Linen Chute Disposal
Locker Bank Staff
Mirror Safety PSY
Mirror Tilting Accessible
Pass Box Isolation
Patient Hoist Ceiling Track
Patient Hoist Mobile
Patient Wardrobe
Sharps Bin Bracket
Shower Cubicle Accessible
Signage Hanging Wayfinding
Signage Queue Display
Signage Room Plate
Signage Totem External
Theatre Panel Modular
Toilet Cubicle Accessible Assembly
Toilet Cubicle Ambulant
Toilet Cubicle HPL
Visitor Lockers Coin
Wall Bumper Bedhead
Wall Panel Hygienic Cladding
Wall Protection Corner Guard
Wall Protection Crash Rail
Wall Protection Handrail Corridor
Wheelchair Rack Wall
Window · Anti Ligature Psychiatric
Window · Door Combination SW
Window · Hatch Cashier
Window · Hatch Pass Through Lab
Window · Hatch Pharmacy Dispensing p.18
Window · Hopper High Level Vent
Window · Laboratory Fixed p.19
Window · Observation Internal Nurse p.17
Window · Observation Theatre Fixed
Window · Pass Hatch Specimen
Window · Screen Reception Glazed
Window · Viewing Isolation
Window · Viewing Lead Glass Xray
Window · Viewing Mortuary
Window · Ward Alu Sliding Restricted
Window · Ward Steel Casement Mesh

Interiors and medical equipment 172

Anaesthesia Rack
Biosafety Cabinet Class II
Breast Pump Station
CSSD Rack Mobile
CSSD Sink Washing SS
CSSD Table SS Work
Clean Utility Unit
Clinical Base Unit SS
Clinical Tall Unit Store
Clinical Wall Unit Glazed
Controlled Drugs Cupboard
Dental Lab Bench
Fume Cupboard
Kitchen Ward Beverage
Kitchen Ward Pantry
Lab Bench Island
Lab Bench Wall
Lead Apron Rack
Nurse Station Counter
Pharmacy Counter Dispensing
Pharmacy Shelving Dispensary
Procedure Room Unit
Reception Counter Accessible
Sluice Unit Base
Sterile Store Shelving Wire
Theatre Storage Wall
Bed Accessory Mattress
Bin Clinical Pedal
Cabinet Records Mobile
Chair Bariatric Waiting
Chair Consulting
Chair Patient Arm
Cot Side Chair Parent
Desk Consulting
Linen Skip
Locker Bedside

All hospital families, continued

Bold names with a page number are shown in this catalogue. Every family carries types, data, keynotes and its service connectors.

Screen Mobile Privacy
Seating Anti Ligature
Seating Corridor Bench
Seating Modular Lounge
Seating Waiting Beam
Sofa Family Room
Step Bed Two Step
Stool Surgical
Table Staff Rest
Table Swaddling
Triage Desk
Analyser Lab Bench
Bath Hoist
Bed Bariatric
Bed Delivery
Bed ICU Electric p.11
Bed Paediatric Cot
Bed Ward Electric
Bed Ward Manual
Birthing Pool
Blood Gas Analyser
Body Trolley Mortuary
Bone Densitometer
C Arm Mobile
CPAP Bubble Neonatal
CSSD Trolley Closed Transport
CT Scanner
CTG Monitor
Cardio Stress Test Station
Cath Lab
Centrifuge Bench
Chair Blood Donor
Chair Dental
Chair Dialysis
Chair Phlebotomy
Chair Recliner Visitor
Chair Weighing
Commode Chair
Cot Infant Bassinet
Couch Examination
Couch Gynaecology
Cylinder O2 Trolley
Defibrillator
Dental Compressor Suction
Dental Lab Device
Diathermy Unit
Dispensing Cabinet Auto
ECG Machine Trolley
ENT Treatment Unit
Endoscope Drying Cabinet
Endoscope Washer
Endoscopy Tower
Flowmeter O2
Fluoroscopy Room
Freezer Minus 80
Fridge Blood Bank
Fridge Vaccine
Gamma Camera
Grossing Station
Incubator Neonatal p.14
Infusion Pump Volumetric
LINAC Bunker
Lab Bench Device Small
Lab Embedding Station
Lab Hot Air Oven
Lab Incubator
Lab Microtome
Lab Platelet Incubator Agitator
Lab Slide Stainer
Lab Tissue Processor
Lab Water Distiller Wall
Light Operating Mobile
Lithotripter ESWL
MRI Scanner
Machine Anaesthesia p.13
Machine Dialysis
Mammography Unit
Microscope Bench
Monitor Patient Mobile
Monitor Patient Wall Arm
Mortuary Body Immersion Tank
Mortuary Fridge 3Body
OPG Unit
Ophthalmic Laser Station
Ophthalmic Operating Microscope
Ophthalmic Perimeter Bench
Ophthalmic Refraction Unit
Oxygen Concentrator
Patient Warming System
Pharmacy Robot
Phototherapy Unit
Physio Bike Static Recumbent
Physio Electrotherapy Trolley
Physio Hot Pack Unit
Physio Mirror Standing
Physio Parallel Bars
Physio Plinth
Physio Quadriceps Bench
Physio Stairs Training
Physio Tilt Table
Physio Traction Table
Physio Treadmill Rehab
Resuscitaire Infant
Scale Adult Height
Scale Infant
Shower Trolley
Slit Lamp Ophthalmic
Stand IV
Suction Unit Wall
Syringe Driver
Table Autopsy
Table Operating
Table Overbed
Tower Endoscopy Types
Trolley Dressing SS
Trolley Instrument Mayo
Trolley Linen
Trolley Meal
Trolley Medicine
Trolley Resuscitation
Trolley Stretcher
UV Disinfection Mobile
Ultrasound Cart
Ventilator ICU
Walking Frame
Warmer Radiant Infant
Warming Cabinet
Water Bath Lab
Wheelchair Bariatric
Wheelchair Standard
Xray Dental Wall
Xray Film Processor Printer
Xray Fixed Room
Xray Mobile
Audiometry Booth
Cold Room Walk In

Mechanical 52

Diffuser Laminar Flow Theatre
Diffuser Linear Slot
Diffuser Square 4Way
Grille Extract Low Level
Grille Return
Grille Transfer Door
HEPA Terminal Filter
Louvre External
Damper Fire
Damper Fire Smoke
Damper Pressure Stabiliser
Damper Volume
Sensor Room Pressure
Sensor Temp Humidity
Thermostat Controller
Valve CAV VAV
AC Cassette 4Way
AC Ducted FCU
AC Precision Server Room
AC Split Outdoor
AC Split Wall High
AHU General

All hospital families, continued

Bold names with a page number are shown in this catalogue. Every family carries types, data, keynotes and its service connectors.

AHU Theatre HEPA
Air Curtain Entrance
Boiler Steam CSSD
Ceiling Fan
Chiller Air Cooled
Condenser Cold Room
Dehumidifier
ERV Fresh Air
Evaporator Cold Room
Extract Fan Fume Cupboard
Extract Fan Inline
Extract Fan Isolation HEPA
Extract Fan Roof
Extract Fan Wall
Fan Coil Cassette Isolation
Fuel Bulk Tank 200h
Fuel Day Tank Bund
Hood Kitchen Extract
Kitchen Cold Room Unit
Kitchen Combi Oven
Kitchen Dishwasher Pass
Laundry Dryer
Laundry Ironer
Laundry Washer Extractor
Medical Waste Autoclave
Pneumatic Tube Blower
Pneumatic Tube Station
Pump Chilled Water
VRF Indoor Cassette
VRF Outdoor

Plumbing and medical gases 85

Med Gas Air Plant
Med Gas Alarm Panel Area
Med Gas Area Valve Box
Med Gas Bedhead Panel
Med Gas Ceiling Column Retractable
Med Gas Ceiling Column Rigid
Med Gas Cylinder Filling Compressor
Med Gas Gas Track Surgical
Med Gas Hose Drop Theatre
Med Gas Manifold O2Cylinder
Med Gas Nitrogen Control Cabinet
Med Gas Outlet AGSS
Med Gas Outlet Combined Rail
Med Gas Outlet Med Air
Med Gas Outlet N2O
Med Gas Outlet O2
Med Gas Outlet Vacuum
Med Gas Oxygen Plant PSA
Med Gas Pendant ICU
Med Gas Pendant Theatre
Med Gas VIE LOX
Med Gas Vacuum Plant
Cleanout Rodding
Floor Drain SS
Floor Gully Sealed Theatre
Grease Interceptor Undersink
Manifold Box Recessed
Manifold Distribution SS
TMV Type 3
Tap Elbow Lever
Tap Sensor
Valve Gate
Valve PRV
Vent Air Admittance
Acid Neutralisation Tank
Autoclave Benchtop
Calorifier
Compressor Oil Free Lab
Drinking Water Dispenser
Ice Machine
Pump Booster Set
Pump Sump Submersible
RO Point Dialysis
Service Point Med Gas
Service Point Water
Steriliser Steam CSSD
Tank Poly Tower
Tank Sectional GRP
Tank Underground Rainwater
UV Steriliser
Ultrasonic Cleaner CSSD
Washer Disinfector CSSD
Washer Glassware Lab
Water Heater Instant
Water Heater Solar
Water Heater Storage
Water Treatment RO Dialysis
Baby Bath
Basin Accessible
Basin Clinical Handwash
Basin Vanity
Bath Assisted
Bath Hydrotherapy Arm Leg
Bedpan Washer Disinfector
Bottle Refill Station
Drinking Fountain Dual Height
Macerator Pulp
Scrub Trough Theatre
Shower Accessible Thermostatic
Shower Decontamination Eye
Shower Emergency Eyewash
Sink Cleaners
Sink Cup Lab
Sink Hopper Sluice Sensor
Sink Kitchen SS Double
Sink Lab Epoxy
Sink Plaster Dental
Sink Plaster Trap
Sink Sluice Hopper
Trough Autopsy Sink
Urinal Bowl
WC Accessible Peninsular
WC Bariatric
WC Squat
WC Standard

Electrical and ELV 71

Comms · PA Speaker Ceiling
Comms · Rack Comms
Comms · Wi Fi Access Point
ATS Panel
Battery Bank
DB Essential Red
DB Final SPN
DB Final TPN
Generator Diesel Standby p.20
Harmonic Filter Active
IPS Theatre IT
Inverter Solar
Isolating Transformer Room Dental
Isolator Rotary
Load Bank Connection Point
MDB Essential
MSB LV Main
Meter Cabinet Intake
Mobile Generator Connection Point
Panel Essential Changeover Room
Transformer Pad
UPS Battery Cabinet
UPS Central
UPS Rack
Voltage Stabiliser Equipment
Access Control Card Reader
Access Control Mag Lock
Bed Exit Sensor Point
Clock Theatre Elapsed
Data Outlet RJ45
Fused Spur
Hand Dryer HEPA
Intercom Door
Isolation Room Pressure Panel
Junction Box
Nurse Call Bedhead Unit
Nurse Call Master Station
Nurse Call Over Door Light
Nurse Call Pull Cord Toilet
RTLS Reader
Socket 13A Essential Red

All hospital families, continued

Bold names with a page number are shown in this catalogue. Every family carries types, data, keynotes and its service connectors.

Socket 13A Twin
Socket Floor Box
Socket Industrial 32A
Socket Medical IT
Surgeon Control Panel
Switch Dimmer
Switch Light
TV Patient Arm
Xray Viewer PACS Display
Batten LED
Bedhead Reading Ward
Bollard Light
Bulkhead IP65
Downlight LED
Emergency Bulkhead Maintained
Examination Mobile
Examination Wall Arm
Exit Sign Illuminated
Floodlight
Linear Corridor Indirect
Night Light Footlight
Operating Light Ceiling p.12
Panel 600 Cleanroom IP65
Panel 600 General
Streetlight Pole
UVC Upper Room
Ward Bedhead Night
Xray Warning Light
Security · CCTV Camera Bullet
Security · CCTV Camera Dome

Fire protection 28

Fire alarm · Beacon Visual Only
Fire alarm · Call Point Manual
Fire alarm · Detector Heat
Fire alarm · Detector Multi
Fire alarm · Detector Smoke
Fire alarm · Door Holder Magnetic
Fire alarm · Interface Module
Fire alarm · Panel Addressable
Fire alarm · Sounder Beacon
Breeching Inlet
Evacuation Mat Cabinet
Extinguisher CO2
Extinguisher Cabinet
Extinguisher Powder
Fire Blanket
Hose Reel Swinging
Hydrant Pillar External
Landing Valve Riser
Refuge Call Point
Signage Fire Action
Suppression Gas Server Room
Valve Alarm Check
Valve Zone Flow Switch
Fire Pump Set p.21
Fire Tank
Sprinkler · Concealed
Sprinkler · Pendent Standard
Sprinkler · Sidewall

Structure 16

Cast In Channel
Embed Plate
Lift Pit
MRI RF Cage
Movement Joint Slab
Opening Radiation Shielded Sleeve
Shielding Lead Panel
Tank Stand Steel
Base Plate
Fin Plate
Base Tank Stand
Pad Footing
Pile Bored
Pile Cap
Plinth Equipment Isolated
Strip Footing

Civil and site 23

Ambulance Canopy Signage
Gatehouse Barrier
Helipad Marking
Incinerator Medical Waste
Kerb Precast
Mortuary Vehicle Bay
Oxygen Compound
Parking Bay Accessible
Parking Bay Ambulance
Speed Hump
Catchpit Stormwater
Culvert Box
Culvert Pipe
Drain U Channel Covered
Grease Trap External
Gully Trapped
Interceptor Fuel
Manhole Brick
Manhole Precast
Neutralisation Pit Lab
Septic Tank Multi Chamber
Soakpit
WWTP Packaged

Landscape 17

Groundcover
Hedge
Palm
Planter Raised Therapy
Shrub
Tree Ornamental
Tree Shade
Bench Concrete Timber Cap
Bench Healing Garden
Drinking Post Outdoor
Fence Chainlink
Gate Sliding Main
Handwash Outdoor
Pergola Timber
Shelter Waiting
Waste Bin External Segregated
Water Feature Healing Garden

Lightning protection and earthing 10

Air Terminal Rod
Conductor Clip
Down Conductor Test Clamp
Earth Bar Main
Earth Pit Inspection
Earth Rod
Equipotential Bar Theatre
Ring Earth Conductor
SPD Type 1
SPD Type 2